

Using R To Analyse An IoT Connected HEALTH MONITORING SYSTEM

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We delve into the deployment of IoT for health monitoring using the statistical programming language R. The primary aim of the system developed here is to harness R's analytical prowess to process and analyse real-time health data acquired from diverse IoT devices.



The health monitoring system proposed here incorporates wearable sensors, mobile applications, and cloud-based infrastructure to establish a seamless connection between individuals and healthcare providers. Wearable devices, equipped with sensors, continuously capture vital health metrics such as heart rate, blood pressure, temperature, and activity levels. These metrics are transmitted to a central server via the internet, where R-based algorithms are utilised for real-time analysis.

R plays a pivotal role in the processing of raw health data, execution of statistical analyses, and generation of meaningful insights. Leveraging the language's extensive

library of statistical functions, visualisation tools, and machine learning capabilities facilitates the development of predictive models aimed at identifying potential health issues. Additionally, R enables the creation of personalised health profiles and trend analyses for individuals, fostering proactive healthcare management. The architecture also encompasses a user-friendly mobile application, providing individuals with real-time access to their health information.

The application described here delivers personalised feedback, health recommendations, and alerts based on the analyses performed by R. Moreover, healthcare providers can remotely monitor

patients' health status and intervene promptly if anomalies are detected.

To ensure the security and privacy of health data, the system incorporates robust encryption and authentication mechanisms. The cloud-based infrastructure, leveraging R, ensures secure data storage and compliance with healthcare data protection regulations.

This proposed IoT-connected health monitoring system exemplifies the substantial contribution R can make to the evolving field of digital healthcare. The objective of this system is to formulate an IoT-based healthcare application that can predict the possibility of a heart attack through smart wearables, using generalised linear